

Osama Ayaz

Full-Stack Web Developer — React, Next.js, Node.js, PostgreSQL
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PROJECTS

- MediAssist** | *Next.js, PostgreSQL, Drizzle ORM, Clerk* Sept. 2025 – Oct. 2025
- Architected and built a full-stack medical voice platform with authenticated users, persistent sessions, and generated clinical summaries
 - Designed relational schemas for users, conversations, transcripts, and reports using Drizzle ORM with PostgreSQL
 - Abstracted third-party speech and AI services behind internal API layers to decouple vendor dependencies
 - Implemented Clerk authentication with middleware-enforced protection across all private routes and API endpoints
 - Improved client-side performance through dynamic imports, route-level code splitting, and optimized data fetching
 - Enabled replayable conversation history and structured report generation for longitudinal user tracking
- Lost & Found Hub** | *Next.js, TypeScript, MongoDB, NextAuth* 2024
- Designed and developed a centralized Lost & Found system to replace fragmented WhatsApp groups and email chains
 - Implemented item reporting, claiming, and verification flows to reduce false ownership and spam
 - Enforced university-only access using email-domain restrictions and middleware-protected routes
 - Built a contributor reputation system with points, badges, and leaderboards to drive user participation
 - Implemented controlled contact sharing where personal details are revealed only after claim validation
 - Designed REST APIs for items, users, claims, and reputation data using MongoDB and Mongoose
- Medify (E-Pharmacy MVP)** | *React, Node.js, Express, PostgreSQL* 2024
- Built a role-based e-pharmacy MVP supporting customers, pharmacies, and administrators with JWT authentication
 - Designed RESTful APIs for medicines, carts, orders, users, and inventory management
 - Implemented real-time order status updates for pharmacies using WebSocket-based notifications
 - Developed cart and checkout workflows with order lifecycle management and pricing calculations
 - Built admin and pharmacy dashboards with analytics visualizations for orders and inventory
 - Applied secure credential handling with bcrypt and enforced authorization at route and controller levels

EXPERIENCE

- Machine Learning Intern** June 2024 – Sept. 2024
Murabbi & Hamsan Tech Islamabad, Pakistan
- Worked on real-world datasets, building complete experimentation and evaluation pipelines
 - Developed internal visualizations and reports to communicate findings to non-technical stakeholders
 - Operated independently in a research-driven engineering environment with minimal supervision
- Flutter Intern** Apr. 2026 – Jun. 2026
Cognitive Health International, NSTP, NUST Islamabad, Pakistan
- Developed and maintained Flutter applications following MVVM architecture with the Stacked framework.
 - Integrated healthcare and IoT devices including Smart Pill Box, Lepod Pro, digital thermometers, and BPW smart watches across Flutter and native Android (Java) implementations.
 - Worked with payment solutions including Stripe, Google Pay, Apple Pay, and Square to support secure transaction workflows.
 - Accelerated device-integration workflows by leveraging AI-assisted development tools, successfully reproducing complex integrations in significantly reduced time.
 - Collaborated with senior engineers in a hybrid development environment, contributing to feature development, debugging, and system integration efforts.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL, HTML, CSS

Frontend: React, Next.js, Tailwind CSS, Flutter

Backend: Node.js, Express, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB

Mobile: Flutter, Dart, Android (Java), Stacked (MVVM)

Auth & Services: Clerk, NextAuth, JWT, Supabase, Neon

Developer Tools: Git, Docker, Vercel, VS Code

Background: Machine Learning, Deep Learning

EDUCATION

National University of Sciences and Technology

Bachelor of Engineering in Software Engineering

Islamabad, Pakistan

Sept. 2023 – May 2027